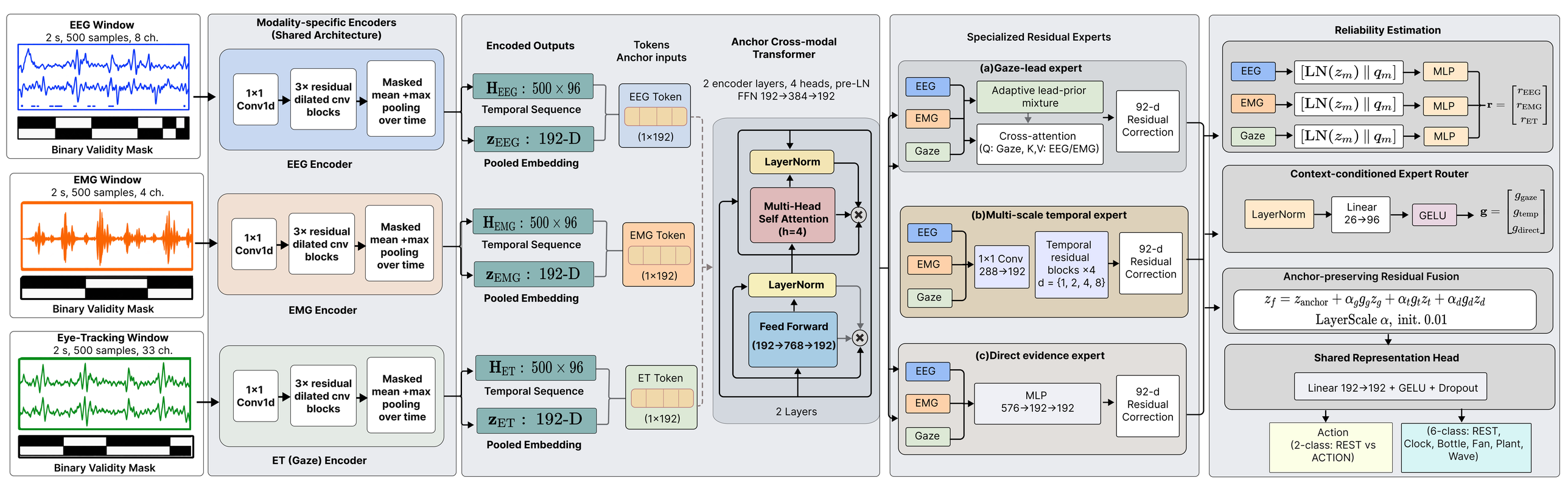
EEG Window
2 s, 500 samples, 8 ch.
Binary Validity Mask
EMG Window
2 s, 500 samples, 4 ch.
Binary Validity Mask
Eye-Tracking Window
2 s, 500 samples, 33 ch.
Binary Validity Mask
Modality-specific Encoders
(Shared Architecture)
1×1 Conv1d
3× residual dilated cnv blocks
Masked mean +max pooling over time
EEG Encoder
EMG Encoder
ET (Gaze) Encoder
Encoded Outputs
Tokens
Anchor inputs
Anchor Cross-modal Transformer
2 encoder layers, 4 heads, pre-LN
FFN 192→384→192
Temporal Sequence
Pooled Embedding
EEG Token
EMG Token
ET Token
(1×192)
LayerNorm
Multi-Head Self Attention (h=4)
Feed Forward (192→768→192)
2 Layers
Specialized Residual Experts
(a)Gaze-lead expert
EEG
EMG
Gaze
Adaptive lead-prior mixture
Cross-attention
(Q: Gaze, K,V: EEG/EMG)
92-d Residual Correction
(b)Multi-scale temporal expert
1×1 Conv 288→192
Temporal residual blocks ×4
d = {1, 2, 4, 8}
(c)Direct evidence expert
MLP 576→192→192
Reliability Estimation
MLP
Context-conditioned Expert Router
LayerNorm
Linear 26→96
GELU
Anchor-preserving Residual Fusion
LayerScale α, init. 0.01
Shared Representation Head
Linear 192→192 + GELU + Dropout
Action
(2-class: REST vs ACTION)
(6-class: REST, Clock, Bottle, Fan, Plant, Wave)